Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current 561 - gemm_kernel_fp32	(16, 16, 1)x(32, 32, 1)	371.49 us	601,763	0 - NVIDIA GeForce RTX 4060 Laptop GPU	1.62 Ghz	[15354] gemm_fp32_cudadma	

Summary | Details | Source | Context | Comments | Raw | Session — Compare | Tools | View | Export

GPU Speed Of Light Throughput

Roofline Single Precision

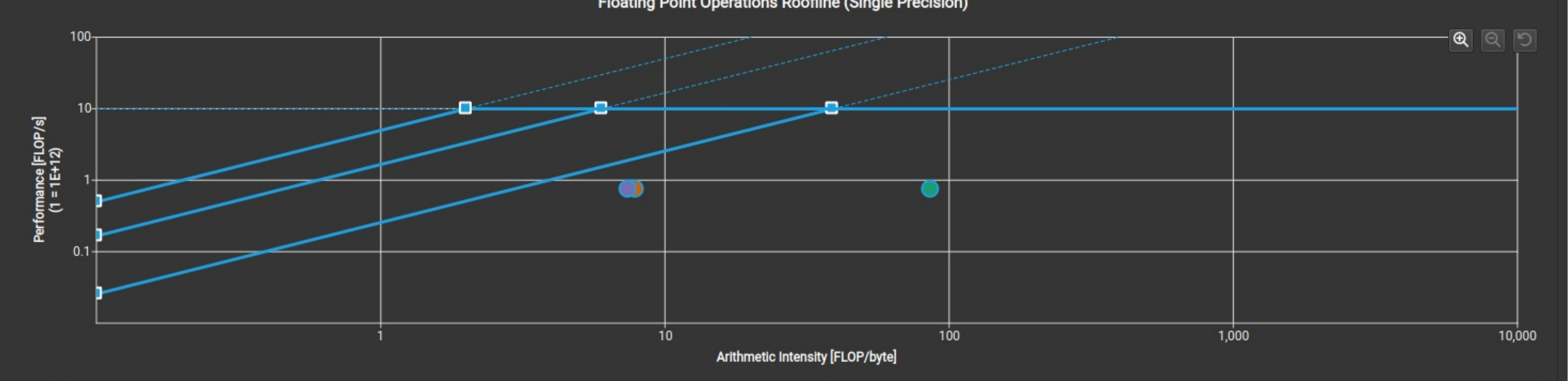
High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	80.32	Duration [us]	371.49
Memory Throughput [%]	80.32	Elapsed Cycles [cycle]	601,763
L1/TEX Cache Throughput [%]	83.35	SM Active Cycles [cycle]	579,855.29
L2 Cache Throughput [%]	5.80	SM Frequency [Ghz]	1.62
DRAM Throughput [%]	3.33	DRAM Frequency [Ghz]	7.99

High Throughput This workload is utilizing greater than 80.0% of the available compute or memory performance of the device. To further improve performance, work will likely need to be shifted from the most utilized to another unit. Start by analyzing workloads in the Compute Workload Analysis section.

Roofline Analysis The ratio of peak float (FP32) to double (FP64) performance on this device is 64:1. The workload achieved 7% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the Profiling Guide for more details on roofline analysis.

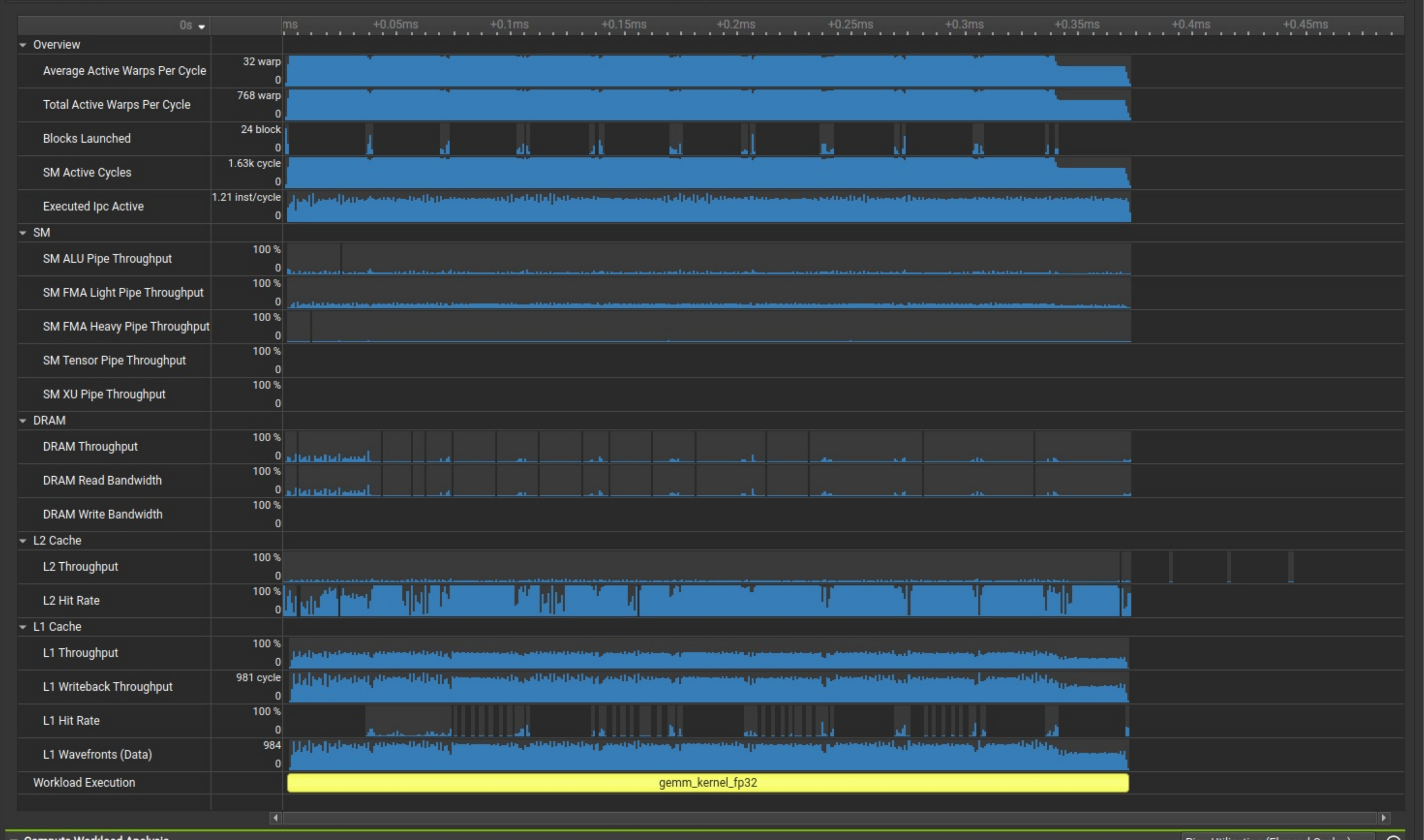
Floating Point Operations Roofline (Single Precision)



PM Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over its runtime.

Maximum Sampling Interval [us]	1	# Pass Groups	2
Maximum Buffer Size [Mbyte]	8.39	-	-



Compute Workload Analysis

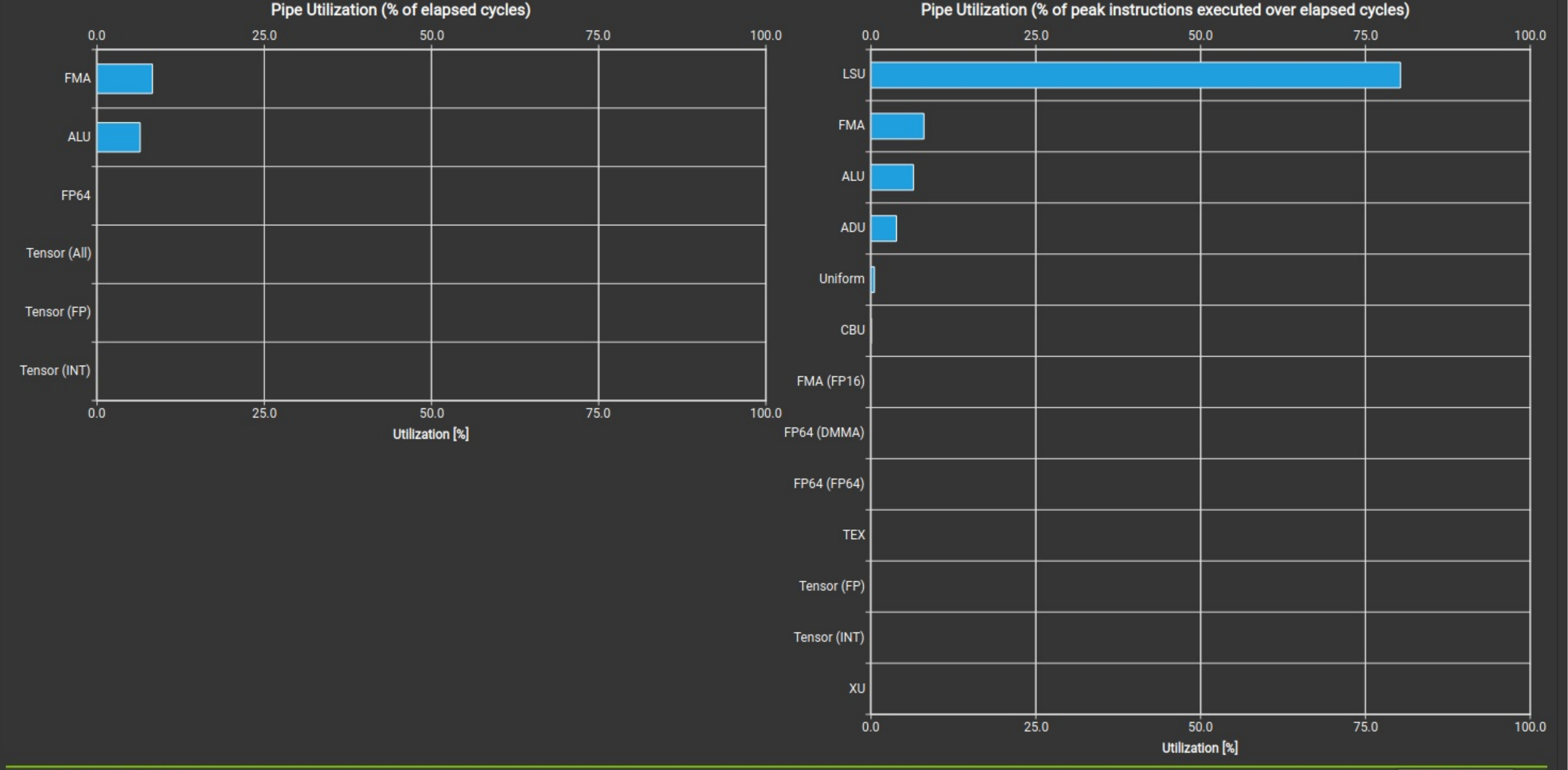
Pipe Utilization (Elapsed Cycles)

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	0.89	SM Busy [%]	28.06
Executed Ipc Active [inst/cycle]	0.93	Issue Slots Busy [%]	22.31
Issued Ipc Active [inst/cycle]	0.93		

Low Utilization
Est. Local Speedup: 91.72% All compute pipelines are under-utilized. Either this workload is very small or it doesn't issue enough warps per scheduler. Check the Launch Statistics and Scheduler Statistics sections for further details.

Key Performance Indicators



Memory Workload Analysis

Memory Chart

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each memory unit.

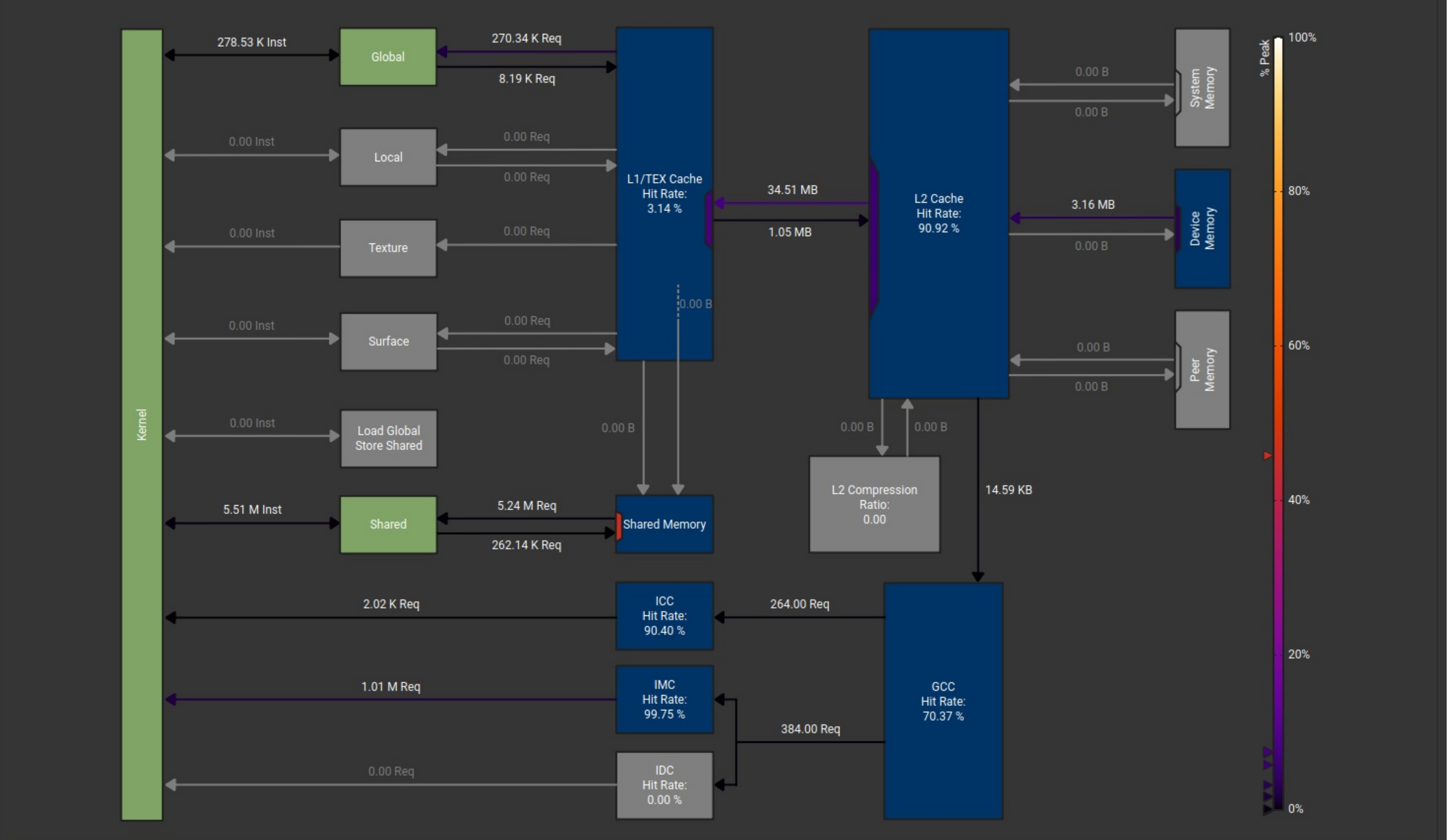
Memory Throughput [Gbyte/s]	8.51	Mem Busy [%]	48.06
L1/TEX Hit Rate [%]	3.14	Max Bandwidth [%]	80.32
L2 Hit Rate [%]	90.92	Mem Pipes Busy [%]	80.32
L2 Compression Input Sectors [sector]	0	Local Memory Spilling Requests	0
L2 Compression Ratio	0	Local Memory Spilling Request Overhead [%]	0
L2 Compression Success Rate [%]	0	L2 Persisting Size [Mbyte]	6.29

Shared Store Bank Conflicts
Est. Speedup: 17.14% The memory access pattern for shared stores might not be optimal and causes on average a 1.3 - way bank conflict across all 262144 shared store requests.This results in 67870 bank conflicts, which represent 20.57% of the overall 330014 wavefronts for shared stores. Check the Source Counters section for uncoalesced shared stores.

Key Performance Indicators

Memory Chart

Values: Transfer Size | Inactivity: Greyed Out



Scheduler Statistics

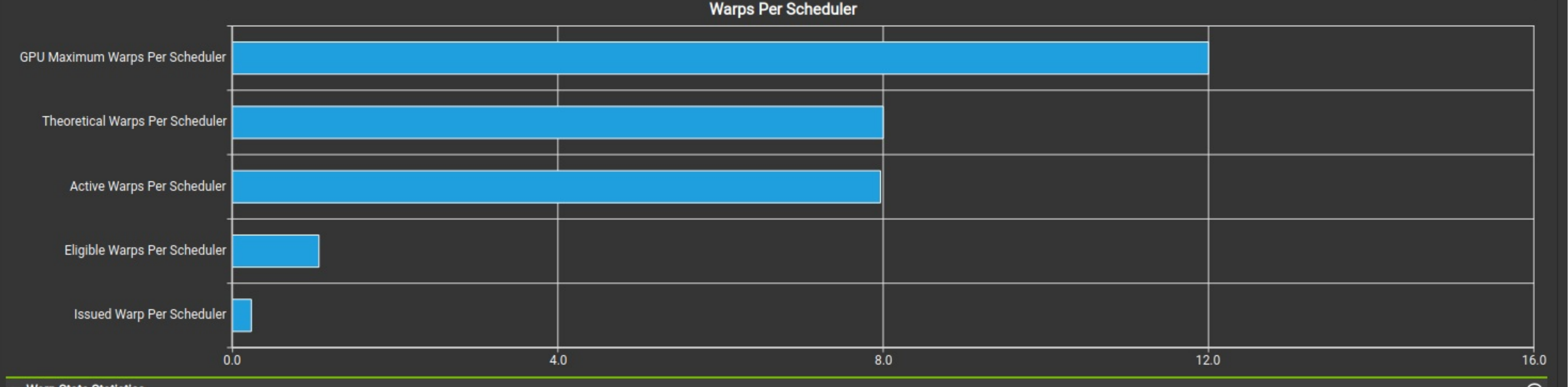
Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Issued Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates poor latency hiding.

Active Warps Per Scheduler [warp]	7.97	No Eligible [%]	76.86
Eligible Warps Per Scheduler [warp]	1.06	One or More Eligible [%]	23.14
Issued Warp Per Scheduler	0.23		

Issue Slot Utilization
Est. Local Speedup: 19.68% Every scheduler is capable of issuing one instruction per cycle, but for this workload each scheduler only issues an instruction every 4.3 cycles. This might leave hardware resources underutilized and may lead to less optimal performance. Out of the maximum of 12 warps per scheduler, this workload allocates an average of 7.97 active warps per scheduler, but only an average of 1.06 warps were eligible per cycle. Eligible warps are the subset of active warps that are ready to issue their next instruction. Every cycle with no eligible warp results in no instruction being issued and the issue slot remains unused. To increase the number of eligible warps, avoid possible load imbalances due to highly different execution durations per warp. Reducing stalls indicated on the Warp State Statistics and Source Counters sections can help, too.

Key Performance Indicators

Warps Per Scheduler



Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

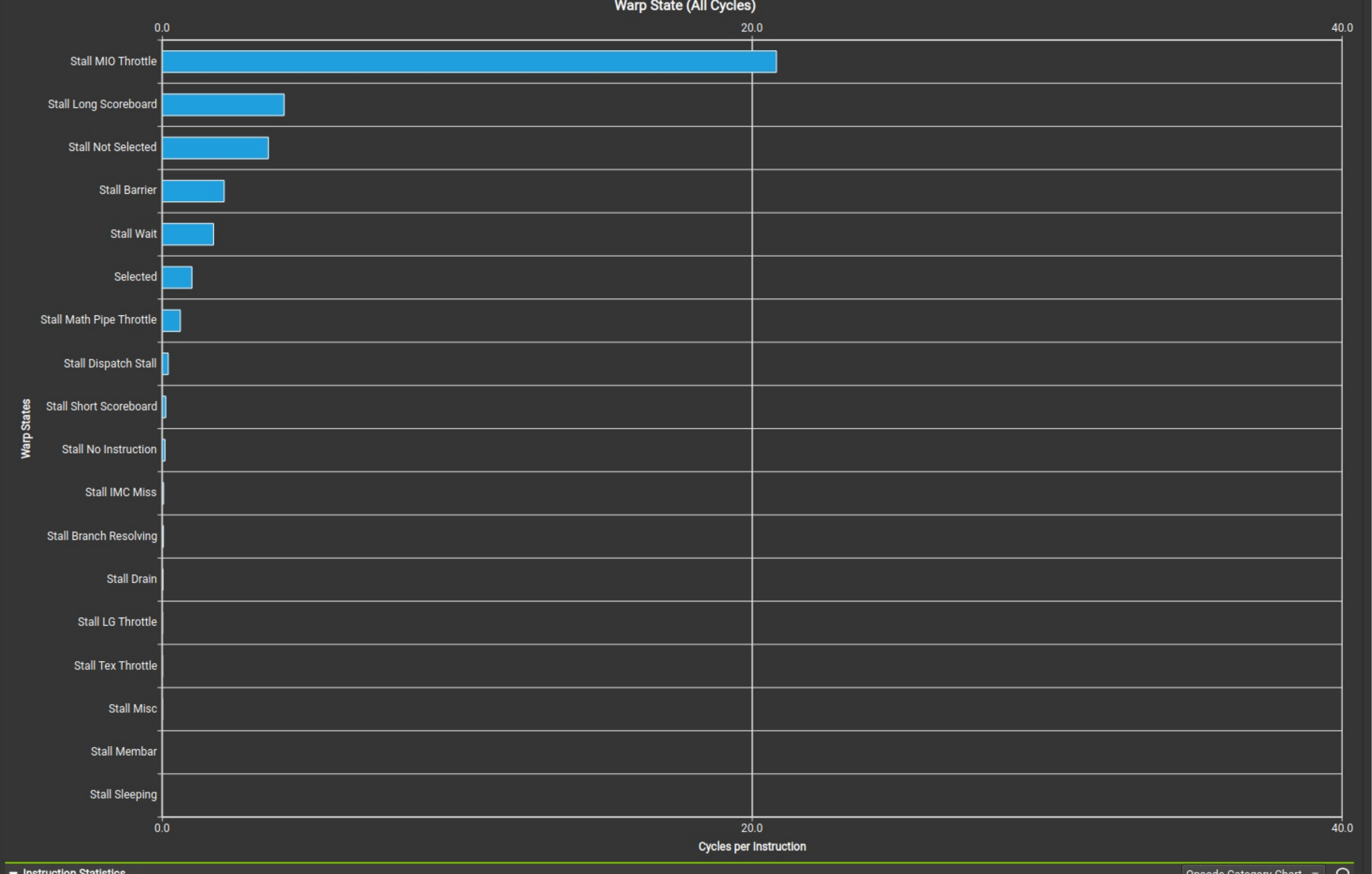
Warp Cycles Per Issued Instruction [cycle]	34.44	Avg. Active Threads Per Warp	32
Warp Cycles Per Executed Instruction [cycle]	34.46	Avg. Not Predicated Off Threads Per Warp	31.94

Mio Throttle Stalls
Est. Speedup: 19.68% On average, each warp of this workload spends 20.8 cycles being stalled waiting for the MIO (memory input/output) instruction queue to be not full. This stall reason is high in cases of extreme utilization of the MIO pipelines, which include special math instructions, dynamic branches, as well as shared memory instructions. When caused by shared memory accesses, trying to use fewer but wider loads can reduce pipeline pressure. This stall type represents about 60.5% of the total average of 34.4 cycles between issuing two instructions.

Key Performance Indicators

Warp Stall Check the Warp Stall Sampling (All Samples) table for the top stall locations in your source based on sampling data. The Profiling Guide provides more details on each stall reason.

Warp State (All Cycles)



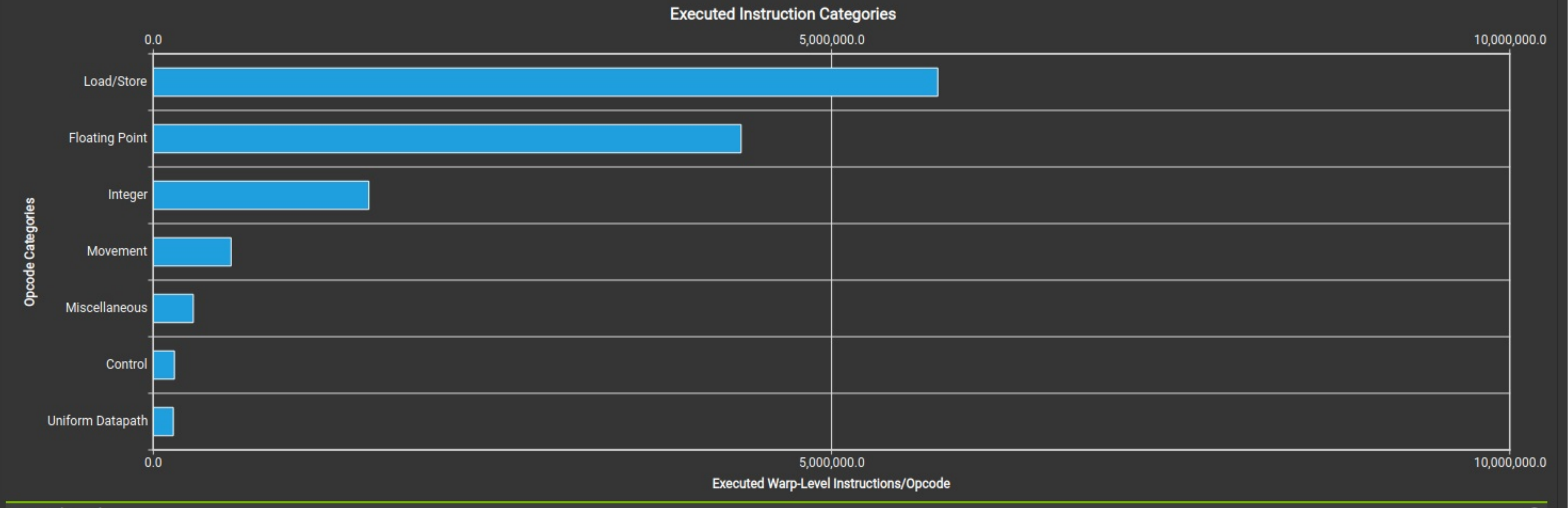
Instruction Statistics

Opcode Category Chart

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	12,877,824	Avg. Executed Instructions Per Scheduler [inst]	134,144
Issued Instructions [inst]	12,887,016	Avg. Issued Instructions Per Scheduler [inst]	134,239.75

Executed Instruction Categories



NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Topology

The system does not have any NVLink connections.

NVLink Tables

Detailed tables with properties for each NVLink.

Logical NVLink Properties

The system does not have any NVLink connections.

NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

NUMA ID Table

GPU ID	GPU Name	NUMA ID by CPU Affinity	CPU Affinity	NUMA ID by Memory Affinity
0	NVIDIA GeForce RTX 4060 Laptop GPU	0	0-15	0

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	256	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	38	Static Shared Memory Per Block [Kbyte/block]	8.19
Block Size	1,024	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	262,144	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	10.67	Shared Memory Configuration Size [Kbyte]	16.38
Uses Green Context	0	Stack Size	1,024
# SMs [SM]	24	# TPCs	12
Enabled TPC IDs	all	-	-

Occupancy

% Occupancy Graphs

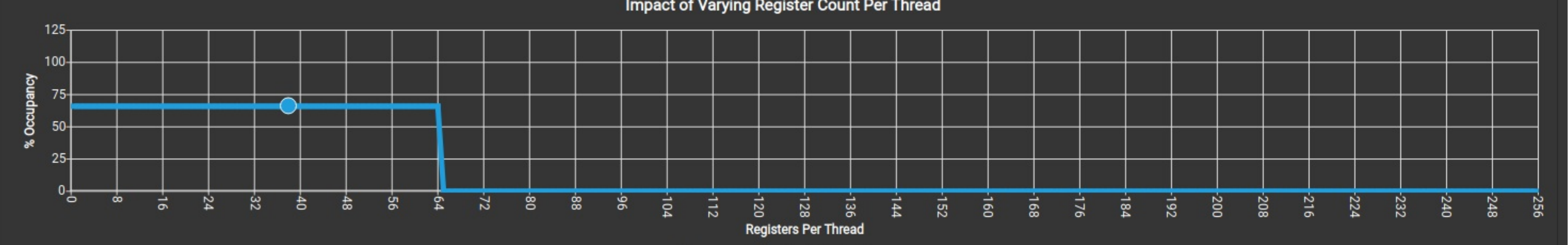
Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	66.67	Block Limit Registers [block]	1
Theoretical Active Warps per SM [warp]	32	Block Limit Shared Mem [block]	1
Achieved Occupancy [%]	66.72	Block Limit Warps [block]	1
Achieved Active Warps Per SM [warp]	32.03	Block Limit SM [block]	24

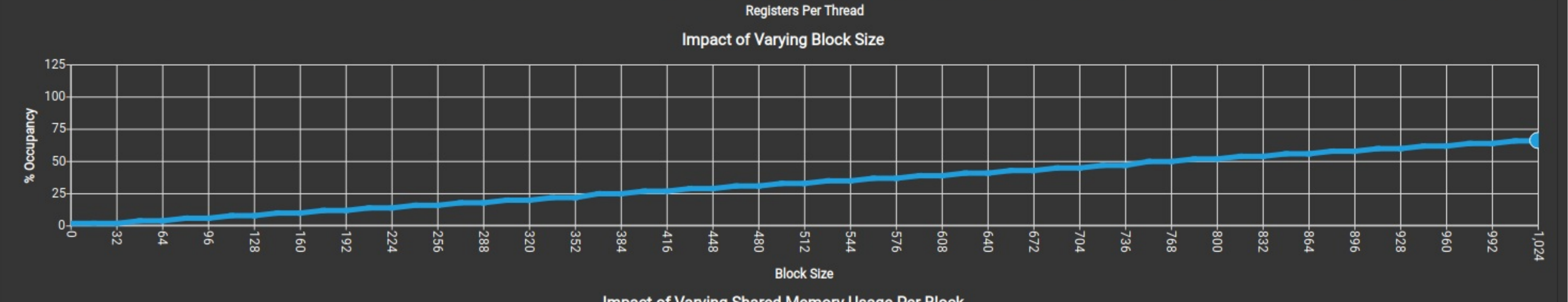
Theoretical Occupancy
Est. Speedup: 19.68% The 8.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (66.7%) is limited by the number of required registers, and the number of warps within each block.

Key Performance Indicators

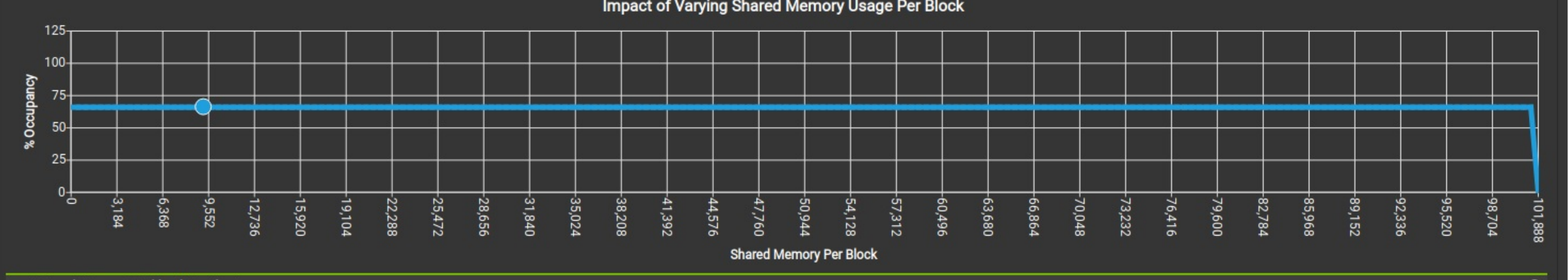
Impact of Varying Register Count Per Thread



Impact of Varying Block Size



Impact of Varying Shared Memory Usage Per Block



GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	579,855.29	Average L1 Active Cycles [cycle]	579,855.29
Average L2 Active Cycles [cycle]	178,665.06	Average SMSP Active Cycles [cycle]	580,233.94
Average DRAM Active Cycles [cycle]	98,828	Total SM Elapsed Cycles [cycle]	14,442,176
Total L1 Elapsed Cycles [cycle]	14,442,176	Total L2 Elapsed Cycles [cycle]	9,626,928
Total SMSP Elapsed Cycles [cycle]	57,768,704	Total DRAM Elapsed Cycles [cycle]	11,876,352

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	579,855.29	545,382	597,768	13,916,527
SMSP Active Cycles	580,233.94	546,028	598,521	55,702,458
L1 Active Cycles	579,855.29	545,382	597,768	13,916,527
L2 Active Cycles	178,665.06	172,457	186,598	2,858,641
DRAM Active Cycles	98,828	98,688	98,976	395,312

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	155,648	Branch Efficiency [%]	100
Branch Instructions Ratio [%]	0.01	Avg. Divergent Branches [branches]	0

Warp Stall Sampling (All Samples)

Location	Value	Value (%)
0x7e9f5f2725a0 in gemm_kernel_fp32	1,401	10
0x7e9f5f272600 in gemm_kernel_fp32	698	5
0x7e9f5f272ab0 in gemm_kernel_fp32	348	3
0x7e9f5f2726c0 in gemm_kernel_fp32	339	2
0x7e9f5f272830 in gemm_kernel_fp32	304	2

Most Instructions Executed

Location	Value	Value (%)
0x7e9f5f272ab0 in gemm_kernel_fp32	131,072	1
0x7e9f5f272aa0 in gemm_kernel_fp32	131,072	1
0x7e9f5f272a90 in gemm_kernel_fp32	131,072	1
0x7e9f5f272a80 in gemm_kernel_fp32	131,072	1
0x7e9f5f272a70 in gemm_kernel_fp32	131,072	1

Follow the *rules outputs* to get guidance on how to navigate through the report and quickly discover performance bottlenecks in this kernel.
You could also disable individual sections to focus on selected performance aspects and make profiling faster.